

we find that optimizing this objective in combination with temperature annealing still minimizes actual variational bounds on validation and test sets. Like the structured output prediction task, we use a multi-sample bound for evaluation with $m = 1000$.

The temperature is annealed using the schedule $\tau = \max(0.5, \exp(-rt))$ of the global training step t , where τ is updated every N steps. $N \in \{500, 1000\}$ and $r \in \{1e-5, 1e-4\}$ are hyperparameters for which we select the best-performing estimator on the validation set and report test performance.

As shown in Figure 4, ST Gumbel-Softmax outperforms other estimators for Categorical variables, and Gumbel-Softmax drastically outperforms other estimators in both Bernoulli and Categorical variables.

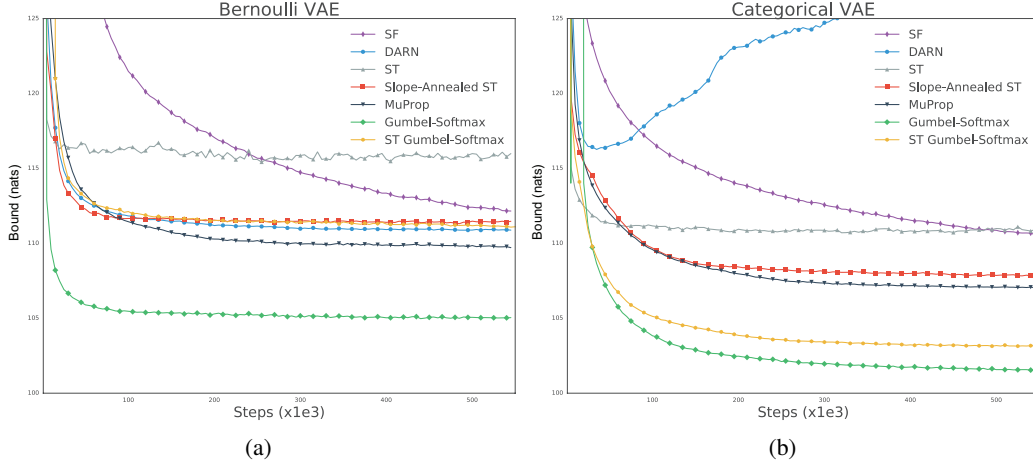


Figure 4: Test loss (negative variational lower bound) on binarized MNIST VAE with (a) Bernoulli latent variables ($784 - 200 - 784$) and (b) categorical latent variables ($784 - (20 \times 10) - 200$).

Table 1: The Gumbel-Softmax estimator outperforms other estimators on Bernoulli and Categorical latent variables. For the structured output prediction (SBN) task, numbers correspond to negative log-likelihoods (nats) of input images (lower is better). For the VAE task, numbers correspond to negative variational lower bounds (nats) on the log-likelihood (lower is better).

	SF	DARN	MuProp	ST	Annealed ST	Gumbel-S.	ST Gumbel-S.
SBN (Bern.)	72.0	59.7	58.9	58.9	58.7	58.5	59.3
SBN (Cat.)	73.1	67.9	63.0	61.8	61.1	59.0	59.7
VAE (Bern.)	112.2	110.9	109.7	116.0	111.5	105.0	111.5
VAE (Cat.)	110.6	128.8	107.0	110.9	107.8	101.5	107.8

4.3 GENERATIVE SEMI-SUPERVISED CLASSIFICATION

We apply the Gumbel-Softmax estimator to semi-supervised classification on the binary MNIST dataset. We compare the original marginalization-based inference approach (Kingma et al., 2014) to single-sample inference with Gumbel-Softmax and ST Gumbel-Softmax.

We trained on a dataset consisting of 100 labeled examples (distributed evenly among each of the 10 classes) and 50,000 unlabeled examples, with dynamic binarization of the unlabeled examples for each minibatch. The discriminative model $q_\phi(y|x)$ and inference model $q_\phi(z|x, y)$ are each implemented as 3-layer convolutional neural networks with ReLU activation functions. The generative model $p_\theta(x|y, z)$ is a 4-layer convolutional-transpose network with ReLU activations. Experimental details are provided in Appendix A.

Estimators were trained and evaluated against several values of $\alpha = \{0.1, 0.2, 0.3, 0.8, 1.0\}$ and the best unlabeled classification results for test sets were selected for each estimator and reported

in Table 2. We used an annealing schedule of $\tau = \max(0.5, \exp(-3e-5 \cdot t))$, updated every 2000 steps.

In Kingma et al. (2014), inference over the latent state is done by marginalizing out y and using the reparameterization trick for sampling from $q_\phi(z|x, y)$. However, this approach has a computational cost that scales linearly with the number of classes. Gumbel-Softmax allows us to backpropagate directly through single samples from the joint $q_\phi(y, z|x)$, achieving drastic speedups in training without compromising generative or classification performance. (Table 2, Figure 5).

Table 2: Marginalizing over y and single-sample variational inference perform equally well when applied to image classification on the binarized MNIST dataset (Larochelle & Murray, 2011). We report variational lower bounds and image classification accuracy for unlabeled data in the test set.

	ELBO	Accuracy
Marginalization	-106.8	92.6%
Gumbel	-109.6	92.4%
ST Gumbel-Softmax	-110.7	93.6%

In Figure 5, we show how Gumbel-Softmax versus marginalization scales with the number of categorical classes. For these experiments, we use MNIST images with randomly generated labels. Training the model with the Gumbel-Softmax estimator is $2\times$ as fast for 10 classes and $9.9\times$ as fast for 100 classes.

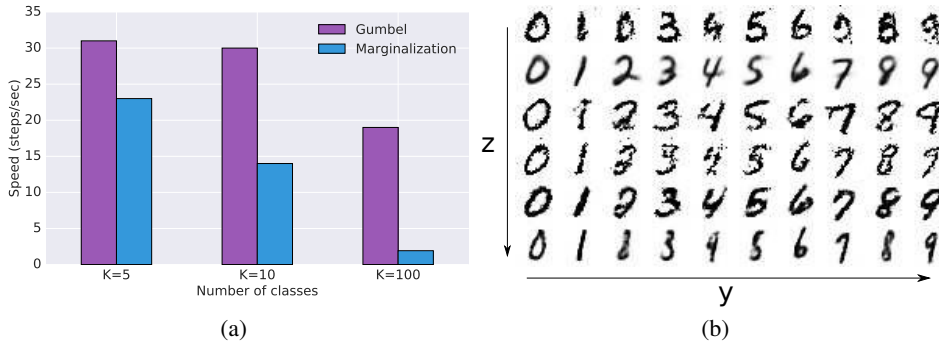


Figure 5: Gumbel-Softmax allows us to backpropagate through samples from the posterior $q_\phi(y|x)$, providing a scalable method for semi-supervised learning for tasks with a large number of classes. (a) Comparison of training speed (steps/sec) between Gumbel-Softmax and marginalization (Kingma et al., 2014) on a semi-supervised VAE. Evaluations were performed on a GTX Titan X[®] GPU. (b) Visualization of MNIST analogies generated by varying style variable z across each row and class variable y across each column.

5 DISCUSSION

The primary contribution of this work is the reparameterizable Gumbel-Softmax distribution, whose corresponding estimator affords low-variance path derivative gradients for the categorical distribution. We show that Gumbel-Softmax and Straight-Through Gumbel-Softmax are effective on structured output prediction and variational autoencoder tasks, outperforming existing stochastic gradient estimators for both Bernoulli and categorical latent variables. Finally, Gumbel-Softmax enables dramatic speedups in inference over discrete latent variables.

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